

Mitigating Effects of Non-ideal Synaptic Device Characteristics for On-chip Learning

Pai-Yu Chen¹, Binbin Lin², I-Ting Wang³, Tuo-Hung Hou³, Jieping Ye², Sarma Vrudhula¹,
Jae-sun Seo¹, Yu Cao¹, and Shimeng Yu^{1*}

¹Arizona State University, Tempe, AZ 85281, USA ²University of Michigan, Ann Arbor, MI 48109, USA ³National Chiao-Tung University, Hsinchu, Taiwan *Email: shimengyu@asu.edu

Abstract—The cross-point array architecture with resistive synaptic devices has been proposed for on-chip implementation of weighted sum and weight update in the training process of learning algorithms. However, the non-ideal properties of the synaptic devices available today, such as the nonlinearity in weight update, limited ON/OFF range and device variations, can potentially hamper the learning accuracy. This paper focuses on the impact of these realistic properties on the learning accuracy and proposes the mitigation strategies. Unsupervised sparse coding is selected as a case study algorithm. With the calibration of the realistic synaptic behavior from the measured experimental data, our study shows that the recognition accuracy of MNIST handwriting digits degrades from ~97 % to ~65 %. To mitigate this accuracy loss, the proposed strategies include 1) the smart programming schemes for achieving linear weight update; 2) a dummy column to eliminate the off-state current; 3) the use of multiple cells for each weight element to alleviate the impact of device variations. With the improved synaptic behavior by these strategies, the accuracy increases back to ~95 %, enabling the reliable integration of realistic synaptic devices in the neuromorphic systems.

Index Terms— machine learning, neuromorphic computing, cross-point array, resistive memory, synaptic device

I. INTRODUCTION

Neuromorphic computing have gained great attention as the traditional Boolean computing based on CMOS technology is reaching its physical limits [1]. The primary goal of neuromorphic computing is to develop bio-inspired artificial systems that enable better interaction with the natural environment for problems such as image recognition, while achieving significant computation speed-up and low power consumption. Several progresses in building large-scale neuromorphic hardware have been made in the recent years, e.g. FACETS [2], Caviar [3], SpiNNaker [4], TrueNorth [5], etc. These systems demonstrated so far are still based on CMOS technology. However, the neural network generally consists of a massive number of synapses that connect between groups of neurons, which requires a large volume of memories and thus results in a tremendous hardware cost [6]. Therefore, it is beneficial to develop more compact synaptic devices based on emerging non-volatile memory devices. Resistive synaptic devices in this paper refer to a special subset of the resistive memory devices that can continuously tune the conductance into multi-level states [7]. To implement the learning algorithm on chip, the conductance of a synaptic device represents a weight element. Prior works suggest that the cross-point array architecture represents the weight matrix and can perform the weighted sum and weight update in a parallel fashion [8-10].

Prior works also addressed the IR drop problem along interconnect wire and its mitigation strategies [11, 12]. In this paper, we focus on the impact of non-ideal properties of synaptic devices, which are rarely discussed in prior works. We also fabricate synaptic devices, and our theoretical study is validated with the experimentally measured data.

An ideal synaptic device behavior assumes a linear update of the weight with the input stimulus, e.g. the voltage pulse. However, in reality, this assumption may not hold. Fig. 1 (a) shows the microscopic top-view image and device structure of our fabricated TaO_x/TiO₂ based synaptic device [13]. Fig. 1 (b) shows that by applying positive (negative) pulses on the device, long-term potentiation (LTP) (long-term depression (LTD)) can increase (decrease) the conductance. Fig. 1 (c) and (d) show other representative LTP/LTD behaviors in the PCMO based [14] and Ag:a-Si based [15] synaptic devices. It is observed that a nonlinear behavior of LTP and LTD commonly exists in today's synaptic devices, possibly due to the inherent drift and diffusion dynamics of the ions/vacancies in these materials.

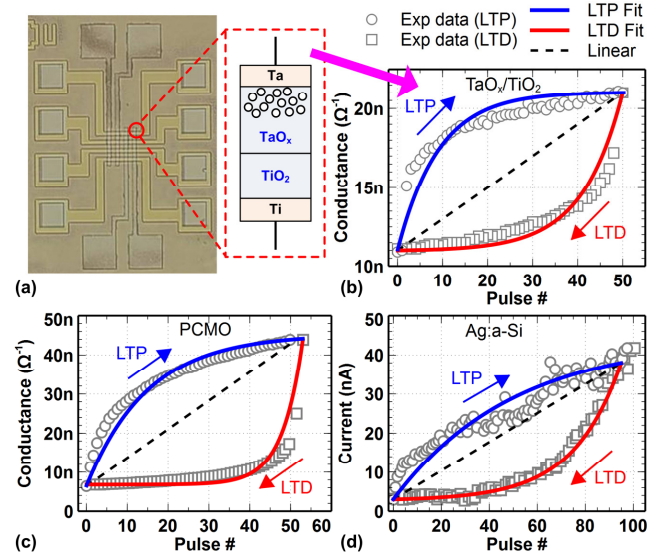


Figure 1 (a) Microscopic top view and device structure of our fabricated TaO_x/TiO₂ based synaptic device [13] and (b) the measured experimental data of conductance modulation for the weight update. Other reported measured experimental data in literature: (c) PCMO [14] and (d) Ag:a-Si [15]. Exponential function is used to fit the LTP and LTD curves.

Besides the *nonlinearity in weight update*, another realistic characteristic deviated from an ideal device is the *limited ON/OFF weight ratio*. The off-state conductance is not perfectly zero in all three examples in Fig. 1, while the ideal

synaptic device assumes that the minimum weight is zero. In addition, the *device variation* is always a concern for the emerging devices at the nanoscale. The synaptic device variation has two aspects, one is the spatial variation from device to device, and the other is the temporal variation from pulse to pulse. All of these realistic device characteristics may degrade the learning performance of neuromorphic systems. Fig. 2 shows a poor recognition accuracy (~65 %) of the MNIST handwriting digits with the realistic synaptic device behavior, compared to the accuracy with ideal device behavior (~97 %) when the sparse coding algorithm [16] is used. Increasing the number of training data sets has limited compensation on the loss of accuracy. In this work, we aim to address these challenges and develop design strategies to mitigate these non-ideal effects of synaptic devices on the learning accuracy. With the proposed strategies in Section IV applied, the recognition accuracy can increase back to ~95 %. The contributions of this work include:

- Developed co-optimization methodologies of the learning algorithm, device characteristics and array architecture for on-chip implementation of neuromorphic network for sparse feature learning.
- Modeled a general behavior of conductance modulation in today's realistic synaptic devices and found out that the limited ON/OFF weight ratio and temporal device variations are critical factors on the learning accuracy degradation, while the impact of the nonlinear weight update and spatial device variations can be partially tolerated by the network's resilience.
- Proposed potential solutions to mitigate the loss of learning accuracy due to realistic device properties. These include a dummy column to eliminate the off-state current, smart programming schemes to improve the linearity of weight update, and the use of multiple cells for each weight element to alleviate the impact of device variations.

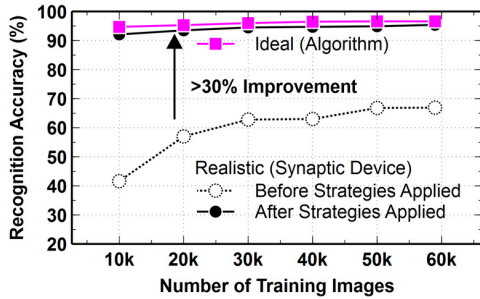


Figure 2 Comparison of the recognition accuracy of the MNIST handwriting digits trained by the sparse coding algorithm using the software approach running and implemented on the hardware architecture with realistic synaptic devices. With the proposed device-level and architecture-level mitigation strategies in Section IV of this work, the recognition accuracy can approach the ideal value of the algorithm.

II. SPARSE CODING ALGORITHM FOR ON-CHIP LEARNING

A. Sparse Coding (SC) Algorithm

Sparse coding algorithm [16] is selected in this work to be implemented with synaptic devices due to its simplicity. Despite of a simple network with two-layer neurons and one weight synaptic matrix, it can still achieve very high learning

accuracy with invariance for pattern's spatial shift and rotation. The sparse coding is found to be a bio-physiological plausible model: neurons in mammalian primary visual cortex can form a sparse representation of natural scenes [17, 18], which is believed to emerge from an unsupervised learning algorithm that attempts to find a factorial code of independent features such as lines, edges and corners. For real-world applications, the sparse coding algorithm has demonstrated its power in numerous domains such as audio processing, text mining and image recognition. In this work, we aim at evaluating and optimizing the synaptic device properties and cross-point architecture for fast and compact on-chip sparse feature learning as a case study.

Fig. 3 (a) shows the simplified process flow of the sparse coding algorithm (SC module) [16]. In the training phase, with a given input vector set $\{X\}$ (braces mean a collection of objects), the corresponding feature vector set $\{Z\}$ and the dictionary matrix (D) are trained iteratively by minimizing the objective error function (E):

$$E = \min \sum (||DZ - X||^2 + \lambda ||Z||_1) \quad (1)$$

As each X is a sparse linear combination of Z via D , the first term of Eq. (1) generally measures how well the dictionary reconstructs the input data. The second term of Eq. (1) imposes constraint of the sparsity of the feature vector. Since both D and Z are unknown, the above optimization problem is a non-convex problem. It is proposed to alternatively optimize Z with fixed D by the coordinate descent (CD) method and optimize D with fixed Z by the stochastic gradient descent (SGD) method, which converts the problem into a convex optimization problem. Using SGD, the D weight update process can be expressed as:

$$D \leftarrow D - \eta RZ^T \quad (R=DZ - X) \quad (2)$$

It can be seen that D is modulated by the product of ηRZ^T , where R is the reconstruction error, and η is the learning rate, which is essentially the delta-rule. For the algorithm ideally implemented in software, the exact value of ηRZ^T can be calculated and applied to the update of D . However, the D update implemented on-chip needs to be translated to the number of pulses applied on the synaptic devices, and the effect of the programming pulses on the conductance of the devices may not represent the exact value of ηRZ^T due to the realistic properties of synaptic devices as mentioned above. In this work, we model the weight update curve and incorporate this model in the D update code in the SC algorithm.

Fig. 3 (b) describes the entire process flow that includes dictionary learning (training phase) and classification (testing phase). In this work, the MNIST handwriting digits [19] is used as the training and testing data set, where the raw images are densely sampled into small patches with 10×10 pixels as X input vector with a dimension of 100. In the later analyses, a set of 40k images is used for training and a different set of 5k images is used for testing. The size of Z is fixed at 300, thus the size of the D matrix is 100×300 ($X \times Z$). After the training process, the trained dictionary D_{train} is used as a fixed D in the testing phase to generate the testing features $\{Z_{\text{test}}\}$. Before the

classification process, a simple maximum pooling operation is employed on both the trained and testing features for each image to select the most active neuron of each feature node, and the output of maximum pooling results in one feature vector per image. Finally, to classify the 10 digits, the support vector machine (SVM) [20] is used. With the input of testing labels, SVM performs classification and gives out the recognition accuracy.

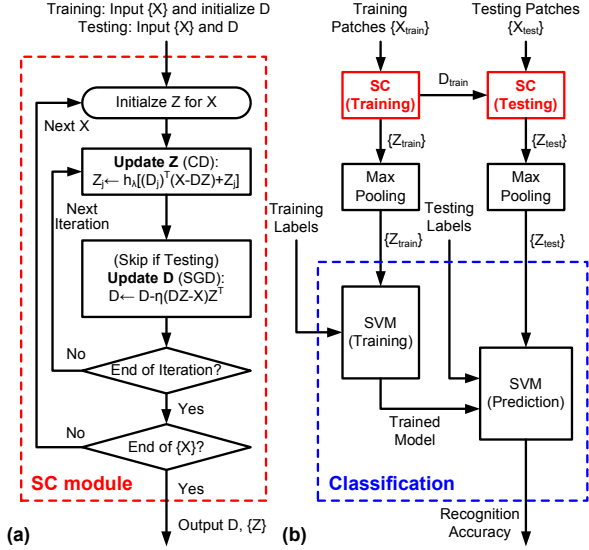


Figure 3 Process flow of (a) the sparse coding (SC) module and (b) the entire process including the unsupervised training by sparse coding (SC) and the supervised classification by support-vector-machine (SVM).

B. Limited On-Chip Precision of SC

To implement the SC algorithm on-chip, it is necessary to limit the precision of D and Z in the algorithm as the chip cannot afford the floating-point computation. In the cross-point architecture, the values in the Z vector are stored on local memories in the peripheral circuitry, and the values in the D matrix are represented by the synaptic weights in the array. Fig. 4 shows the learning accuracy with different precisions by truncation of the bits in the SC algorithm. It suggests that a 4-bit Z is sufficient for high learning accuracy and limited precision of D has more impact on the accuracy. Since the number of bits D is related to how many levels of conductance that the synaptic device can achieve, a 6-bit D (64 levels) is chosen for later analysis based on the number of multi-level states available in today's synaptic devices (see Fig. 1).

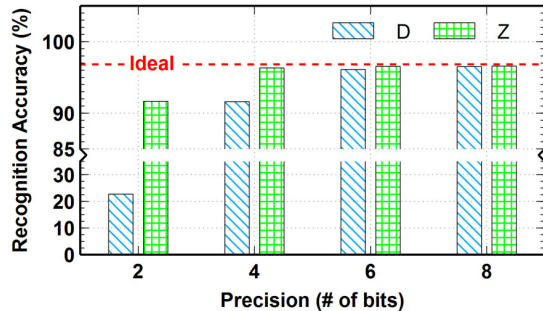


Figure 4 Learning accuracy with different precision bits of D and Z in the SC algorithm.

III. SYNAPTIC DEVICE PROPERTIES

As previously shown in Fig. 1, realistic synaptic behaviors include 1) the nonlinearity in weight update, 2) limited ON/OFF weight ratio, and 3) device variations. In this section, these realistic behaviors are modeled individually into the sparse coding algorithm and their impact on the learning accuracy is investigated. As a baseline, the limited precision of the synaptic devices (64 levels) is considered.

A. Nonlinear Weight Update

To analyze the impact of nonlinear weight update on the learning, a general behavior that models the conductance change of LTP (G_{LTP}) and LTD (G_{LTD}) with the number of pulses (P) is described with the following equations:

$$G_{LTP} = B \left(1 - e^{-\frac{P}{A}} \right) + G_{min} \quad (3)$$

$$G_{LTD} = -B \left(1 - e^{-\frac{P-P_{max}}{A}} \right) + G_{max} \quad (4)$$

$$B = \frac{G_{max} - G_{min}}{1 - e^{-\frac{P_{max}}{A}}} \quad (5)$$

where G_{max} , G_{min} and P_{max} are directly extracted from the experimental data, which represents the maximum conductance, minimum conductance and the maximum pulse number required to switch the device between the minimum and maximum conductance states. A is the parameter that controls the nonlinear behavior of the weight update, and B is simply a function of A that fits the functions within the range of G_{max} , G_{min} and P_{max} . A and B may be different in (3) and (4). A set of nonlinear LTP and LTD behavior can be obtained by adjusting A as illustrated in Fig. 5 (a), where each nonlinear curve is labeled with a nonlinearity value from +6 to -6. Here the plus and minus are merely the signs to label LTP and LTD, respectively.

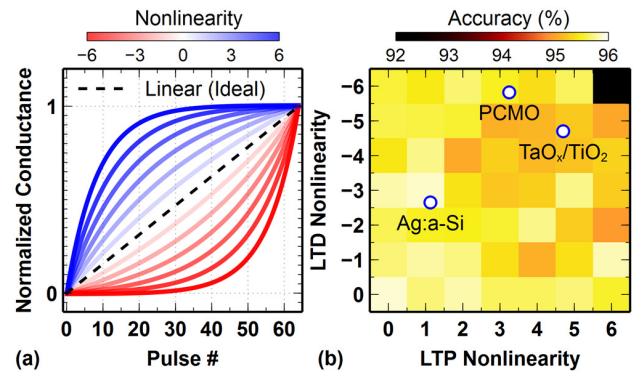


Figure 5 (a) Different nonlinearities of the LTP and LTD fit from +6 to -6. (b) Learning accuracy with different LTP and LTD nonlinearities. Nonlinearities of the three realistic synaptic devices from Fig. 1 are also shown.

Then, we apply these nonlinear functions into the weight update in the SC algorithm. Fig. 5 (b) shows that learning accuracy slightly decreases in the high nonlinearity region of LTP and LTD, and a relatively larger drop from ~96% to ~92% occurs at maximum nonlinearities (+6/-6 curves). For today's synaptic devices (in Fig. 1), the nonlinearities of LTP and LTD

are also labeled in the Fig. 5 (b). It is shown that the nonlinearity in the weight update has a moderate impact on the learning performance.

B. Limited ON/OFF Weight Ratio

Ideally the D values in the SC algorithm are represented by a normalized conductance of synaptic devices, and the range of the D value is from 0 to 1. However, the minimum conductance can be regarded as $D=0$ only when the ratio between the maximum and minimum conductance (ON/OFF ratio) approaches infinity, which is not feasible in today's synaptic devices. Fig. 6 shows the learning accuracy with different ON/OFF ratios. The learning accuracy dramatically decreases when the ON/OFF ratio shrinks below 25, because the calculations involved with small values of D in the algorithm will be significantly distorted. The Ag:a-Si device exhibits a largest ON/OFF ratio of ~ 15 among the devices in Fig. 1, while other devices show even smaller ON/OFF ratio. This means that without any optimization, none of these synaptic devices can lead to high recognition accuracy when used in on-chip implementation of sparse learning.

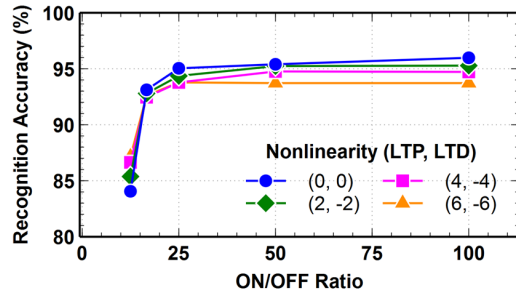


Figure 6 Learning accuracy with different ON/OFF ratios. Four curves are shown for the nonlinearity baselines from (0,0) to (6,-6). The synaptic devices in Fig. 1 have a maximum ratio about 15, which results in $>10\%$ accuracy loss.

C. Device Variations

It is well known that the synaptic devices involving drift and diffusion of the ions/vacancies show considerable variation from device to device (also known as spatial variation), and even from pulse to pulse within one device (also known as temporal variation). Owing to the spatial variation, different devices in the array will follow different nonlinearity baselines. Owing to the temporal variation, there will be pulse to pulse noise on top of the nonlinearity baseline.

1) Spatial Variation of Weight Update

The effect of spatial variation can be analyzed by introducing the variation into the nonlinearity baseline for each synaptic device, as illustrated in Fig. 7. For example, if a synaptic device has a $+100\%$ spatial variation, there will be a $+1$ deviation of the nonlinearity. As shown in Fig. 8 (a), the learning accuracy is insignificantly affected by the spatial variation even with 30% standard deviation from the baseline.

2) Temporal Variation of Weight Update

The temporal variation of the conductance occurs at every weight update operation on the synaptic device, as illustrated in Fig. 7. In this work, the temporal variation is defined as the variation of weight change with one applied voltage pulse. As

shown in Fig. 8 (b), with larger temporal variation, the learning accuracy degrades further due to the disturbance of variation on the weight update process.

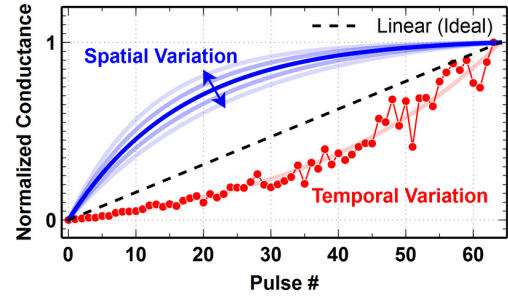


Figure 7 Illustration of the spatial and temporal variation in the weight update. Owing to the spatial variation, different devices in the array will follow different nonlinearity baselines. Owing to the temporal variation, there will be pulse to pulse noise on top of the nonlinearity baseline.

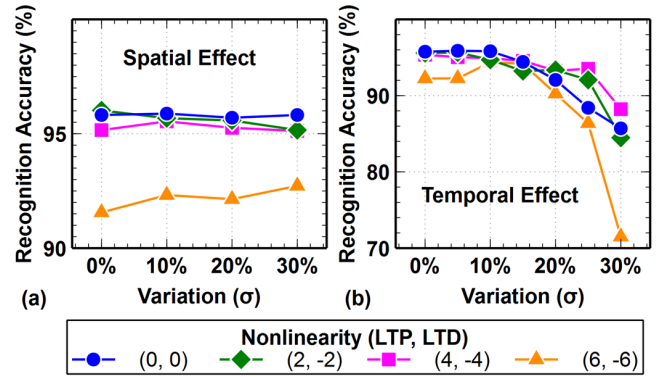


Figure 8 Learning accuracy with different standard deviations of (a) spatial and (b) temporal device variations. The impact of the temporal variation may be critical to the learning accuracy, while the impact of the spatial variation is much less significant.

The results above show that the impact of the temporal variation may be critical to the learning accuracy, while the impact of the spatial variation as well as the nonlinear weight update is much less significant. The reason may be that the SC algorithm can partially tolerate the spatial variation or the nonlinear weight update as the solution to the optimized D is not unique as this is a non-convex problem. With the skewed weight update, the optimized D may have converged at the next local minima. As long as such skew in weight update is deterministic and moderate, the network has the resilience to static variations in weight update. However, if there is substantial temporal variation, which is stochastic in nature, then the network has very little resilience to such non-deterministic disturbance.

IV. PROPOSED STRATEGIES TO IMPROVE LEARNING ACCURACY

In this section, several strategies are proposed to mitigate the non-ideal effects of synaptic devices, including 1) the smart programming scheme for achieving linear weight update; 2) a dummy column to eliminate the off-state current; 3) the use of multiple cells for each weight element to alleviate the impact of device variations. Our study shows that with the improved synaptic behavior by these strategies, the recognition accuracy

increases from ~65 % to ~95 %, enabling reliable integration of realistic synaptic devices in neuromorphic systems.

A. Dummy Column for Elimination of OFF-state Current

It was previously shown that a low ON/OFF ratio of synaptic devices can lead to a significant degradation on the learning accuracy and a large ON/OFF ratio is not available in today's synaptic devices. Instead, one approach to remedy this situation is to eliminate the effect of the off-state current in every weight element with the aid of a dummy column. The cross-point array architecture with a dummy column is illustrated in Fig. 9. The synaptic devices in the dummy column remain in their minimum conductance states, such that the readout value at the output of dummy column represents the weighted sum of the Z vector and the off-state conductance. In the peripheral circuitry, we subtract the off-state weighted sum from all the partial weighted sums, $D_i Z$, performed along the columns. Except for spatial variation between the synaptic devices in the same row, this virtually eliminates the effect of off-state current in the sparse learning task. An additional column will give 1% overhead on the array area as there are totally 100 columns ($X=100$), which is tolerable since the cross-point array is integrated on top of its CMOS peripheral circuitry at interconnect layers.

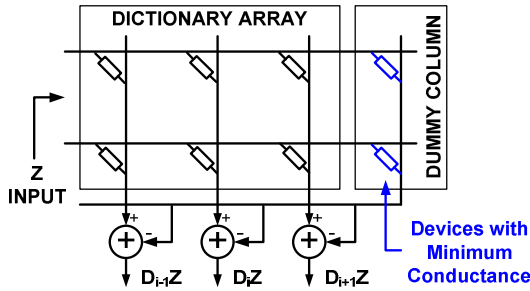


Figure 9 Cross-point array architecture with the dummy column and subtractors to eliminate the common off-state current by differential readout circuits.

B. Programming Schemes for Linear Weight Update

From the device engineering's point of view, it might not be feasible for today's synaptic devices to achieve super-linear conductance modulation when applying a number of voltage pulses. Alternatively, the programming scheme that updates the weight can be smartly designed to improve the linearity. Fig. 10 (a) shows different programming schemes and Fig. 10 (b) shows the corresponding measured experimental weight update in our $\text{TaO}_x/\text{TiO}_2$ based synaptic device. Scheme A uses a simple pulse train for both the potentiation and depression with identical pulses, which leads to the largest nonlinearity. Scheme B is a reinforcement of Scheme A, which splits a single voltage pulse into a pair of positive and negative pulse with different amplitude and duration. It improves the linearity as the second negative pulse can cancel some overshoot of first positive pulse at the beginning stages of LTP and LTD. Scheme C is a more complicated extension of Scheme A, where the pulse duration varies depending on the current conductance state of the synaptic device. The idea of Scheme B and C is to slow down the weight update at the beginning stages of LTP and LTD. It is observed that with identical pulses, the conductance changes very rapidly at the beginning stages of the LTP or LTD and then it gradually saturates. The duration of programming pulses can

be adjusted in a way that a shorter pulse is applied at the beginning stages while gradually wider pulses are applied at subsequent stages. For this scheme, an empirical function to determine the pulse duration (P_D) is required to program the device from conductance state n to $n+1$ ($n=0$: minimum (maximum) conductance for LTP (LTD)) is expressed as:

$$P_D(n \rightarrow n+1) = P_i \times e^{\left(\frac{\max n}{50}\right)} \quad m = \begin{cases} 6, & \text{LTP} \\ 4, & \text{LTD} \end{cases} \quad (6)$$

where P_i is the initial pulse duration.

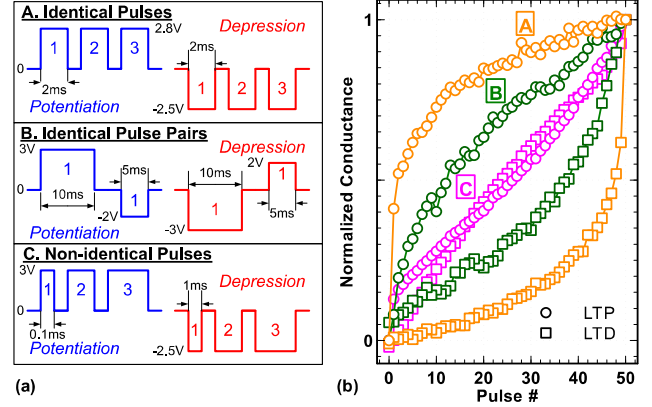


Figure 10 (a) Different programming schemes and (b) the corresponding experimentally measured data of conductance modulation in the $\text{TaO}_x/\text{TiO}_2$ based synaptic device.

We can see that with Scheme C, the weight update approaches the linear curve. However, the trade-off is that Scheme C needs a read-before-write step to first identify the conductance state and then apply the correct pulse duration to the device, which inevitably increases the complexity of the peripheral circuitry design as well as the latency and energy consumption. In particular, Scheme C cannot be applied with the parallel weight update scheme [12] due to non-identical pulses. Instead, the weight matrix can only be updated in a sequential row-by-row fashion. In contrast, Scheme B, although it uses a pulse pair as one programming operation, it does not need the read-before-write step as the pulse pair shape is independent of the conductance state. For quick estimation on the overhead of the array, the latency and energy consumption are simply calculated by the applied pulse widths and voltage amplitudes in these schemes. Assuming that the sparse Z vector has 5% nonzero elements, a LTD weight update from the 30th back to 20th weight level gives ~7.5X latency for Scheme B and ~60X for Scheme C. The overhead of energy consumption will be a bit less than that of latency as the RMS values of voltage in Scheme B and C are smaller. From peripheral circuit design's point of view, identical pulse pairs of Scheme B can be realized by two pulse generators at the two ends of the array to generate a pair of pulses with different polarities. However, Scheme C requires at least an extra computing unit to calculate (6) and a special pulse generator to produce non-identical pulses with fine-grained duration. As shown in Section III. A, since the algorithm has resilience to the moderate weight nonlinearity, Scheme B may be a better choice for a practical implementation considering the overhead of Scheme C.

C. Redundant Cells to Suppress Device Variations

To alleviate the impact of device variations, we propose using multiple cells as one D weight element. This approach statistically averages out the conductance variations of synaptic devices both spatially and temporally. If n cells are used as one weight element, the standard deviation of variations will be reduced by a factor of $1/\sqrt{n}$ assuming that variations are normally distributed. Fig. 11 shows an example of the reduction on the variation using 9 cells compared to that using only 1 cell. This strategy is believed to have considerable improvement on the accuracy loss due to device variations. However, using multiple cells inevitably increases the area of cross-point array as well as the energy consumption by n times. The upper limit on the number of the redundant cells might be constrained by area of the peripheral circuitry since the synaptic devices can be integrated at the interconnect levels. We estimate that the cross-point array area with 9 cells is still smaller than that of the peripheral circuitry in 65 nm [12].

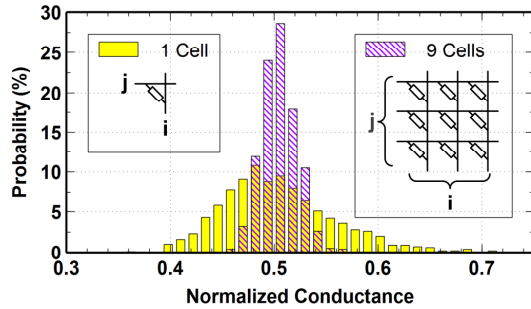


Figure 11 Illustration of multiple cells as one weight element to average out the device variations.

D. Accuracy Improvement by Proposed Strategies

Finally, we revisit the issue in Section I where the realistic synaptic devices cause degradation on the learning accuracy. We implemented the proposed strategies into the SC algorithm. Specifically, it is assumed that the following improvements on the realistic properties are achieved: 1) the ON/OFF weight ratio is increased by 4X from 12.5 (within the range of the Ag:a-Si device) to 50, using a dummy column but assuming that the off-state current is not completely removed due to spatial variation; 2) the nonlinearity reduces from (+6, -6) to (+3, -3) if Scheme B of the smart programming is applied; 3) 9 cells as a weight element is used to reduce both spatial and temporal variation from 20 % (for the Ag:a-Si device) to 6.7 %. As shown in Fig. 2 at the beginning of this paper, the recognition accuracy of synaptic devices can closely approach that of the ideal algorithm, achieving an accuracy improvement of >30%.

V. CONCLUSION

Synapses are the core elements of a neuromorphic system to establish communication between groups of neurons. Synaptic devices available today exhibit non-ideal device properties, e.g., the nonlinearity in weight update, limited ON/OFF range and device variations. Sparse coding algorithm is used to provide a platform to evaluate the performance of unsupervised learning using synaptic devices for image applications. It is found that the realistic synaptic device properties can lead to significant degradation on image recognition accuracy from ~97 % to ~65

%. The strategies to remedy this issue are then proposed, including 1) a dummy column to eliminate the off-state current; 2) the smart programming schemes to achieve linear weight update; 3) the use of multiple cells for each weight element to alleviate the impact of device variations. By applying these strategies with tolerable trade-offs on chip area, latency and energy, the synaptic behavior is greatly improved and the recognition accuracy returns to ~95 %, viably enabling the synaptic devices for practical hardware implementation of bio-inspired neuromorphic computing systems.

VI. ACKNOWLEDGEMENT

This work is in part supported by NSF-CCF-1449653.

REFERENCES

- [1] J. Hasler, et al., "Finding a roadmap to achieve large neuromorphic hardware systems," *Frontiers in neuroscience*, vol. 7, 2013.
- [2] J. Schemmel, et al., "A wafer-scale neuromorphic hardware system for large-scale neural modeling," *IEEE International Symposium on Circuits and Systems (ISCAS)*, pp. 1947-1950, 2010.
- [3] R. Serrano-Gotarredona, et al., "CAVIAR: A 45k neuron, 5M synapse, 12G connects/s AER hardware sensory-processing-learning-actuating system for high-speed visual object recognition and tracking," *IEEE Transactions on Neural Networks*, vol. 20, no. 9, pp. 1417-1438, 2009.
- [4] X. Jin, et al., "Modeling spiking neural networks on SpiNNaker," *Computing in Science & Engineering*, vol. 12, no. 5, pp. 91-97, 2010.
- [5] P. A. Merolla, et al., "A million spiking-neuron integrated circuit with a scalable communication network and interface," *Science*, vol. 345, no. 6197, pp. 668-673, 2014.
- [6] J. Partzsch, et al., "Analyzing the scaling of connectivity in neuromorphic hardware and in models of neural networks," *IEEE Transactions on Neural Networks*, vol. 22, no. 6, pp. 919-935, 2011.
- [7] D. Kuzum, et al., "Synaptic electronics: materials, devices and applications," *Nanotechnology*, vol. 24, no. 38, pp. 382001, 2013.
- [8] M. Hu, et al., "Memristor crossbar based hardware realization of BSB recall function," *International Joint Conference on Neural Networks (IJCNN)*, pp. 1-7, 2012.
- [9] P. Gu, et al., "Technological exploration of rram crossbar array for matrix-vector multiplication," *Asia and South Pacific Design Automation Conference (ASP-DAC)*, pp. 106-111, 2015.
- [10] D. Querlioz, et al., "Bioinspired networks with nanoscale memristive devices that combine the unsupervised and supervised learning approaches," *International Symposium on Nanoscale Architectures (NANOARCH)*, pp. 203-210, 2012.
- [11] B. Liu, et al., "Reduction and IR-drop compensations techniques for reliable neuromorphic computing systems," *International Conference on Computer-Aided Design (ICCAD)*, pp. 63-70, 2014.
- [12] P.-Y. Chen, et al., "Technology-design co-optimization of resistive cross-point array for accelerating learning algorithms on chip," *Design, Automation & Test in Europe Conference & Exhibition (DATE)*, pp. 854-859, 2015.
- [13] I.-T. Wang, et al., "3D synaptic architecture with ultralow sub-10 fJ energy per spike for neuromorphic computation," *International Electron Devices Meeting (IEDM)*, pp. 665-668, 2014.
- [14] S. Park, et al., "Neuromorphic speech systems using advanced ReRAM-based synapse," *IEEE International Electron Device Meeting (IEDM)*, pp. 625-628, 2013.
- [15] S. H. Jo, et al., "Nanoscale memristor device as synapse in neuromorphic systems," *Nano letters*, vol. 10, no. 4, pp. 1297-1301, 2010.
- [16] H. Lee, et al., "Efficient sparse coding algorithms," *Advances in neural information processing systems*, pp. 801-808, 2006.
- [17] D. H. Hubel, et al., "Receptive fields, binocular interaction and functional architecture in the cat's visual cortex," *The Journal of physiology*, vol. 160, no. 1, pp. 106-154, 1962.
- [18] A. J. Bell, et al., "The 'independent components' of natural scenes are edge filters," *Vision research*, vol. 37, no. 23, pp. 3327-3338, 1997.
- [19] Y. LeCun, et al., "The MNIST database of handwritten digits," 1998.
- [20] C.-C. Chang, et al., "LIBSVM: a library for support vector machines," *ACM Transactions on Intelligent Systems and Technology (TIST)*, vol. 2, no. 3, pp. 27, 2011.